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United States Patent	12418997
Kind Code	B2
Date of Patent	September 16, 2025
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Display device

Abstract

The present disclosure is related to a display device. The display device includes a middle frame, a display panel and a supporting member. The middle frame is provided with a first mating surface. The display panel includes a display substrate and a cover plate. A first notch is provided at a side of an edge of the cover plate close to the first mating surface. The first notch extends along a thickness direction of the cover plate. An orthographic projection of the display substrate on the cover plate at least partially overlaps an orthographic projection of the first mating surface on the cover plate. The supporting member includes a first supporting section and a second supporting section, the first supporting section is connected to the first mating surface, and the second supporting section is partially arranged in the first notch.

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Appl. No.:	18/028242
Filed (or PCT Filed):	May 12, 2022
PCT No.:	PCT/CN2022/092502
PCT Pub. No.:	WO2023/216189
PCT Pub. Date:	November 16, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240306324 A1	Sep. 12, 2024

Publication Classification

Int. Cl.: H05K5/02 (20060101); H05K5/03 (20060101)

U.S. Cl.:

CPC H05K5/0217 (20130101); H05K5/03 (20130101);

Field of Classification Search

CPC: H05K (5/0217); H05K (5/03); G06F (1/163); G06F (1/1626); G02F (1/133308); G02F (1/133317); G02F (1/1333)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application is based on PCT filing PCT/CN2022/092502, filed May 12, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the display technical field, and in particular, to a display device.

BACKGROUND

(3) At present, according to market demands, more and more consumers have increasingly high requirements on the frame of display devices, and pursue extremely narrow bezel.

(4) When an existing display device is assembled, a relatively wide connection surface is reserved on a cover plate to connect the display panel with a middle frame, resulting in a large frame of the display device.

(5) It should be noted that the information disclosed in the background section above is only used to enhance the understanding of the background of the present disclosure, and therefore may include information that does not constitute the prior art known to those skilled in the art.

SUMMARY

(6) The present disclosure aims at providing a display device to overcome the shortcomings in the above-mentioned related arts.

(7) According to an aspect of the present disclosure, there is provided a display device, including a display panel, a middle frame and a supporting member.

(8) The middle frame is provided with a first mating surface.

(9) The display panel includes a display substrate and a cover plate arranged on a side of the display substrate. A first notch is provided at a side of an edge of the cover plate close to the first mating surface, the first notch extends along a thickness direction of the cover plate, and an orthographic projection of the display substrate on the cover plate at least partially overlaps an orthographic projection of the first mating surface on the cover plate.

(10) The supporting member includes a first supporting section and a second supporting section which are connected sequentially, wherein the first supporting section is connected to the first mating surface, and the second supporting section is partially arranged in the first notch.

(11) In an example of the present disclosure, a thickness of the second supporting section is less than or equal to a distance between a side surface of the first notch and a side surface of the cover

plate.

(12) In an example of the present disclosure, a second notch is provided at a side of the first notch, the second notch extends perpendicular to a thickness direction of the cover plate, and the supporting member further includes a third supporting section which is connected to an end away from the first supporting section, and the third supporting section is snap-fitted in the second notch.

(13) In an example of the present disclosure, the first supporting section is spaced apart from the display substrate.

(14) In an example of the present disclosure, the display substrate has a display region and a binding region located at a periphery of the display region, and the orthographic projection of the first mating surface on the cover plate at least partially overlaps an orthographic projection of the binding region on the cover plate.

(15) In an example of the present disclosure, a width of the first supporting section is greater than or equal to a width of the first mating surface, and the orthographic projection of the binding region on the cover plate is located in an orthographic projection of the first supporting section on the cover plate.

(16) In an example of the present disclosure, a width of the third supporting section is smaller than a width of the first supporting section.

(17) In an example of the present disclosure, the middle frame is provided with an installation notch, the cover plate is arranged in the installation notch, and the first mating surface is an end surface of the installation notch; and the middle frame further includes a second mating surface, the second mating surface is a side surface of the installation notch, and the first notch and the second notch are arranged at a position of the side surface of the cover plate which is opposite to the second mating surface.

(18) In an example of the present disclosure, the supporting member is attached to the first side surface and divided into at least two parts along an extending direction of the first side surface.

(19) In an example of the present disclosure, a flexible buffer layer is provided between the third supporting section and the second notch.

(20) In an example of the present disclosure, the supporting member is made of stainless steel.

(21) In an example of the present disclosure, a third notch is provided at the second mating surface, and the supporting member further includes a fourth supporting section, and the fourth supporting section is snap-fitted in the third notch.

(22) In an example of the present disclosure, the second notch extends continuously along an extending direction of an edge of the cover plate.

(23) In an example of the present disclosure, the second notch includes a plurality of recesses, and the plurality of recesses are distributed at intervals along the extending direction of the side surface of the cover plate.

(24) In an example of the present disclosure, the edge of the cover plate is arc-shaped, and a height of the edge of the cover plate away from the first mating surface is gradually decreased from an end away from close to a center of the cover plate to an end away from the center.

(25) In an example of the present disclosure, the first supporting section is adhered to the first mating surface.

(26) According to another aspect of the present disclosure, there is provided a display device, including the display panel according to another aspect of the present disclosure.

(27) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments consistent with the disclosure and together with the description serve to explain the principles of the disclosure. Apparently, the drawings in the following description are only some embodiments of the present disclosure, and those skilled in the art can obtain other drawings according to these drawings without creative efforts.
- (2) FIG. 1 is a cross-sectional structure diagram of a display panel involved in an embodiment of the present disclosure.
- (3) FIG. 2 is a cross-sectional structure diagram of a display substrate involved in an embodiment of the present disclosure.
- (4) FIG. 3 is a partial schematic diagram of a display device involved in an embodiment of the present disclosure.
- (5) FIG. 4 is a partial schematic diagram of another display device involved in an embodiment of the present disclosure.
- (6) FIG. 5 is a partial schematic diagram of another display device involved in an embodiment of the present disclosure.
- (7) FIG. 6 is a schematic diagram showing cooperation between a supporting member and a cover plate of another display device involved in an embodiment of the present disclosure.
- (8) FIG. 7 is a partial schematic diagram of another display device involved in an embodiment of the present disclosure after the width of a first supporting section is increased.
- (9) FIG. 8 is a three-dimensional structure diagram of another display device involved in an embodiment of the present disclosure.
- (10) FIG. 9 is a partial schematic diagram of another display device involved in an embodiment of the present disclosure.
- (11) FIG. 10 is a partial schematic diagram of another display device involved in an embodiment of the present disclosure.
- (12) FIG. 11 is a cross-sectional view along a first direction of another circular display device involved in an embodiment of the present disclosure.
- (13) FIG. 12 is a cross-sectional view along a first direction of another rectangular display device involved in an embodiment of the present disclosure.

DETAILED DESCRIPTION

- (14) Example embodiments will now be described more fully with reference to the accompanying drawings. However, the example embodiments can be implemented in various forms and should not be construed as limited to the examples set forth herein; rather, providing these embodiments makes the present disclosure more comprehensive and complete, and conveys the concepts of the example embodiments comprehensively to those skilled in the art. The same reference numerals in the drawings denote the same or similar structures, and thus their repeated descriptions will be omitted. Furthermore, the drawings are merely schematic illustrations of the present disclosure and are not necessarily drawn to scale.
- (15) Although relative terms such as “upper” and “lower” are used in the specification to describe a relative relationship of one component shown in the drawings to another component, these terms are used in the specification only for convenience, and are for example based on the direction(s) as illustrated in the drawings. It will be appreciated that if the illustrated device is turned over so that it is upside down, then a component described as being “upper” will become an element that is “lower”. When a structure is “on” another structure, it may mean that the structure is integrally formed on another structure, or that the structure is “directly” placed on another structure, or that the structure is “indirectly” placed on another structure through other structure(s).
- (16) The terms “a”, “an”, “the”, “said” and “at least one” are used to indicate the presence of one or more elements/components/etc. The terms “includes/include” and “have/has” are used to indicate

open-ended inclusion and to mean that additional elements/components/etc. may exist in addition to the listed elements/components/etc. The terms “first”, “second” and “third” are used as markers only and are not quantitative restrictions on their associated objects.

(17) As shown in FIG. 1, an existing display device includes a display panel and a middle frame 8. The display panel includes a display substrate 1. The display substrate 1 includes a display portion 101, a first bending portion 102, and a binding portion 103. The display portion 101 and the binding portion 103 are adjacent to both sides of the first bending portion 102, the binding portion 103 is adjacent to a side of the first bending portion 102 away from the display portion 101 and is bonded, through the first bending portion 102, to a side of the display portion 101 which is the back of the display surface. The display portion 101 may perform the display function of the display panel, and there is a display device inside the display portion 101. The binding portion 103 is provided with a circuit board 2, and the circuit board 2 is connected with metal traces, and the metal traces extend through the first bending portion 102 to the display portion 101 and connected to the display device to realize the binding with the binding portion 103.

(18) It should be noted that the display panel may be an LCD display panel or an OLED display panel.

(19) The display device generally adopts an OLED display panel. As shown in FIG. 2, an OLED display panel generally includes a driving backplane 10, a planarization layer group 14, and a pixel layer 16. The driving backplane 10 may include a base substrate 11 and a driving circuit layer. The driving circuit layer is arranged on a side of the base substrate 11. The planarization layer group 14 is arranged on a side of the driving circuit layer away from the base substrate 11. The pixel layer 16 is arranged on a side of the planarization layer group 14 away from the base substrate 11.

(20) The OLED display panel may further include a first buffer layer 12, and the first buffer layer 12 is arranged between the base substrate 11 and the driving circuit layer.

(21) The base substrate 11 may be a base substrate 11 of inorganic material, or may be a base substrate 11 of organic material. For example, in one embodiment of the present disclosure, the material of the base substrate 11 may be a glass material such as soda-lime glass, quartz glass, sapphire glass, or may be a metallic material such as stainless steel, aluminum, nickel, etc.

(22) In another embodiment of the present disclosure, the base substrate 11 may also be a flexible base substrate. For example, the material of the base substrate 11 may be polyimide (PI). The base substrate 11 may also be a composite of multiple layers of materials. For example, in one embodiment of the present disclosure, the base substrate 11 may include a bottom film, a pressure-sensitive adhesive layer, a first polyimide layer and a second polyimide layer which are stacked sequentially.

(23) In embodiments of the present disclosure, the driving circuit layer is provided with driving circuits for driving sub-pixels. In the driving circuit layer, any one of the driving circuits may include a transistor and a storage capacitor. Further, the transistor may be a thin film transistor 13, and the thin film transistor 13 may be selected from a top-gate thin film transistor, a bottom-gate thin film transistor or a double-gate thin film transistor. Taking a top-gate thin film transistor as an example, the thin film transistor 13 may include an active layer 131, a gate insulating layer 132, a gate electrode 133, a first electrode 136 and a second electrode 137.

(24) The active layer 131 is arranged on a side of the substrate 11. The material of the active layer 131 may be amorphous silicon semiconductor material, low temperature polysilicon semiconductor material, metal oxide semiconductor material, organic semiconductor material or other types of semiconductor materials. The thin film transistor 13 may be an N-type thin film transistor or a P-type thin film transistor. The active layer 131 may include a channel region and two doped regions of different doping types located on both sides of the channel region.

(25) The gate insulating layer 132 may cover the active layer 131 and the base substrate 11. The material of the gate insulating layer 132 is an insulating material such as silicon oxide.

(26) The gate electrode 133 is arranged on a side of the gate insulating layer 132 away from the

base substrate **11**, and is directly facing the active layer **131**. That is, the projection of the gate electrode **133** on the base substrate **11** is located within the projection range of the active layer **131** on the base substrate. For example, the projection of the gate electrode **133** on the base substrate **11** coincides with the projection of the channel region of the active layer **131** on the base substrate **11**. (27) The thin film transistor **13** further includes an interlayer insulating layer **134**. The interlayer insulating layer **134** covers the gate electrode **133** and the gate insulating layer **132**. The thin film transistor **13** further includes an interlayer dielectric layer **135**. The interlayer dielectric layer **135** is arranged on a side of the interlayer insulating layer **134** away from the base substrate **11**. Both the interlayer insulating layer **134** and the interlayer dielectric layer **135** are insulating materials, but the materials of the interlayer insulating layer **134** and the interlayer dielectric layer **135** may be different.

(28) The first electrode **136** and the second electrode **137** are arranged on a surface of the interlayer dielectric layer **135** away from the base substrate **11**. The first electrode **136** may be a first source electrode, and the second electrode **137** may be a drain electrode. The first electrode **136** and the second electrode **137** are connected to the active layer **131**. For example, the first electrode **136** and the second electrode **137** are respectively connected to two corresponding doped regions of the active layer **131** through via holes.

(29) A protection layer **138** is provided on a side of the first electrode **136** away from the base substrate **11**. The protection layer **138** covers the first electrode **136** and the second electrode **137**. A planarization layer group **14** is provided on a side of the first electrode **136** and the second electrode **137** away from the base substrate **11**. The planarization layer group **14** is arranged on a side of the protective layer **138** away from the base substrate **11**. The planarization layer group **14** covers the protection layer **138**, and a surface of the planarization layer group **14** away from the base substrate **11** is a flat surface.

(30) Specifically, the planarization layer group **14** may include a first planarization layer **141** covering the protection layer **138**. The display panel may further include a third electrode **139**. A second planarization layer **142** is provided on a side of the third electrode **139** away from the base substrate **11**. The second planarization layer **142** covers the first electrode **136**, the second electrode **137** and a side of the first planarization layer **141** away from the base substrate **11**. The third electrode **139** is connected to the first electrode **136** through a via hole. It should be noted that the third electrode **139** may be a second source electrode.

(31) On a side of the planarization layer group **14** away from the base substrate **11**, a pixel defining layer **15** and a pixel layer may be provided. The pixel defining layer **15** has a plurality of pixel openings **151**. The pixel layer **16** includes a plurality of sub-pixels, and the plurality of sub-pixels are respectively arranged inside the plurality of pixel openings **151**. An array of plurality of sub-pixels is distributed on a side of the driving backplane **10** away from the base substrate **11**. Specifically, sub-pixels may be located on a side of the planarization layer group **14** away from the base substrate **11**. It should be noted that the sub-pixels may include red sub-pixels, green sub-pixels and blue sub-pixels according to different luminescent colors.

(32) The pixel layer **16** may include a plurality of pixel electrodes **161**, a light-emitting layer **162** and a common electrode **163**. The pixel electrode **161** is located on the surface of the driving backplane **10** away from the base substrate **11**, and the light-emitting layer **162** is arranged on the surface of the pixel electrodes **161** away from the base substrate **11**, and the common electrode **163** is arranged on the surface of the light-emitting layer **162** away from the base substrate **11**.

(33) A pixel electrode **161** is connected to the first electrode **136** or the third electrode **139**. When the thin film transistor **13** only includes the first electrode **136** and the first planarization layer **141**, the pixel electrode **161** is connected to the first electrode **136** through a via hole in the first planarization layer **141**, and the pixel defining layer **15** is set to cover the first electrode **161**, the first planarization layer **141**. When the thin film transistor **13** further includes the third electrode **139** and the second planarization layer **142**, the pixel electrode **161** is connected to the third

electrode **139** through the via hole in the second planarization layer **142**, and the pixel defining layer **15** is set to cover the third electrode **139** and the second planarization layer **142**.

(34) The common electrode **163** may be used as a cathode, and the pixel electrode **161** may be used as an anode. The pixel electrode **161** is connected to a positive electrode of a power supply, and the common electrode **163** is connected to a negative electrode of the power supply. Signals may be applied through the pixel electrode **161** and the common electrode to drive the light-emitting layer **162** to emit light, so as to display images. The specific light-emitting principle will not be described in detail here. The light-emitting layer **162** may include an electroluminescent organic light-emitting material. For example, the light-emitting layer **162** may include an auxiliary layer and a light-emitting material layer sequentially stacked on the pixel electrode **161**. Generally, a pattern region is provided on a mask plate, and the auxiliary layers of sub-pixels of different colors and the light-emitting layers **162** of sub-pixels of different colors are formed by evaporation and other processes.

(35) In addition, the display panel in embodiments of the present disclosure may further include an encapsulation layer **17**, which is arranged on a side of the pixel layer **16** away from the base substrate **11**, so as to encapsulate the pixel layer **16** and prevent water and oxygen erosion. The encapsulation layer **17** may be a single-layer or multi-layer structure, and the material of the encapsulation layer **17** may include organic or inorganic materials, which are not specifically limited here.

(36) In an embodiment, the encapsulation layer **17** may include a first inorganic encapsulation layer **171**, an organic encapsulation layer **172** and a second inorganic encapsulation layer **173**. The first inorganic encapsulation layer **171** is arranged on a side of the pixel layer **16** away from the base substrate **11**. The organic encapsulation layer **172** is arranged on a side of the first inorganic encapsulation layer **171** away from the base substrate **11**. The second inorganic encapsulation layer **173** is arranged on a side of the organic encapsulation layer **172** away from the base substrate **11**.

(37) As shown in FIG. 1, a supporting layer group **3** is provided between the display portion **101** and the binding portion **103** of the display substrate **1**. In order to closely bond the display substrate **1** and the supporting layer group **3** to obtain a good display effect, the profiled one end of the display substrate **1** is bonded to one side of the supporting layer group **3**, and the other end of the display substrate **1** is bonded to the opposite side of the supporting layer group **3**. Specifically, the supporting layer group **3** has a first surface and a second surface oppositely arranged. The display substrate **1** includes a first straight portion, a first bent portion **102** and a second straight portion connected in sequence. The first straight portion is bonded to the first surface of the supporting layer group **3**, and the second straight portion is bonded to the second surface of the supporting layer group **3**.

(38) The supporting layer group **3** includes a first back film **31**, a supporting block **32**, a supporting plate **33**, a first adhesive layer **34**, a second buffer layer **35**, a second adhesive layer **36** and a second back film **37** arranged in sequence. The supporting block **32** is arranged on a side of the first back film **31**. The supporting plate **33** is arranged on a side of the supporting block **32** away from the first back film **31**. The first adhesive layer **34** is arranged on a side of the supporting plate **33** away from the first back film **31**. The second buffer layer **35** is arranged on a side of the first adhesive layer **34** away from the first back film **31**. The second adhesive layer **36** is arranged on a side of the second buffer layer **35** away from the first back film **31**. The second back film **37** is arranged on a side of the second adhesive layer **36** away from the first back film **31**. In order to ensure the supporting strength, the materials of the supporting plate **33** and the supporting block **32** may be metal materials.

(39) It should be noted that the first surface of the supporting layer group **3** refers to a surface of the second back film **37** away from the first back film **31**, and the second surface of the supporting layer group **3** refers to the other surface which is directly opposite to the surface of the first back film **31**. The first straight portion is bonded to one side of the second back film **37** away from the

first back film **31**, and the second straight portion is bonded to the other side opposite to the one side of the first back film **31**. Specifically, a part of the base substrate **11** located at the binding portion **103** is attached to the first back film **31**, and a part of the base substrate **11** located at the display portion **101** is attached to the second back film **37**.

(40) A polarizer **4** is provided on a side of the display substrate **1** away from the display portion **101**. Specifically, a part of the encapsulation layer **17** located in the display portion **101** is attached to the polarizer **4**. A tactile control layer may also be provided on a side of the encapsulation layer **17** located in the display portion **101** away from the supporting layer group **3**. The polarizer **4** is provided on a side of the tactile control layer away from the supporting layer group **3**. The side of the polarizer **4** away from the display portion **101** is provided with a third adhesive layer **5**. A side of the third adhesive layer **5** away from the display portion **101** is bonded to the cover plate **6**. The cover plate **6** can be adhered to the display substrate **1** mounted on its lower surface, and can effectively protect the display substrate **1**.

(41) As shown in FIG. 3, when the display panel is connected to a middle frame **8**, a first mating surface **811** is provided on the middle frame **8**, and a relative wide connection surface is reserved on a side of the cover plate **6** close to the middle frame **8**. The third adhesive layer **5** is provided between the connection surface and the first mating surface **811** to adhere the connection surface and the first mating surface **81**, thus assembling the cover plate **6** and the middle frame **8** together. When this method is used for connection, the width of the reserved connection surface is usually relatively large, resulting in a large bezel of the display device.

(42) In view of the above, embodiments of the present disclosure provide a display device. As shown in FIG. 4 to FIG. 9, the display device includes a display panel, a middle frame **8** and a supporting member **9**. The middle frame **8** is provided with a first mating surface **811**. The display panel includes a display substrate **1** and a cover plate **6** arranged on a side of the display substrate **1**. A first notch **61** is provided at a side of an edge of the cover plate **6** close to the first mating surface **811**. The first notch **61** extends along the thickness direction of the cover plate **6**. The orthographic projection of the display substrate **1** on the cover plate **6** and the orthographic projection of the first mating surface **811** on the cover plate **6** at least partially overlap. The supporting member **9** includes a first supporting section **91** and a second supporting section **92** connected sequentially. The first supporting section **91** is connected to the first mating surface **811**. The second supporting section **92** is partially arranged in the first notch **61**.

(43) The middle frame **8** is provided with the first mating surface **811**. The first notch **61** is provided at the side of the edge of the cover plate **6** close to the first mating surface **811**. The supporting member **9** includes the first supporting section **91** and the second supporting section **92** connected sequentially. The first supporting section **91** is connected to the first mating surface **811**, and the second supporting section **92** is partially arranged in the first notch **61**. The cover plate **6** and the middle frame **8** are connected together through the supporting member **9**, and the first mating surface **811** is moved inwardly to be under the display substrate **1**, so that the orthographic projection of the first mating surface **811** of the middle frame **8** on the cover plate **6** at least partially overlaps the orthographic projection of the display substrate **1** on the cover plate **6** to reduce the distance between the display substrate **1** and a side of a side surface of the middle frame **8** away from the cover plate **6**, thereby reducing the frame of the display device.

(44) As shown in FIG. 4, a side surface of the middle frame **8** close to the display panel is in a stand orientation, the middle frame **8** is provided with an installation notch **81**, and the display panel is arranged in the installation notch **81**. The installation notch **81** includes the first mating surface **811** and the second mating surface **812**. The first mating surface **811** is an end surface of the installation notch **81**, and the second mating surface **812** is a side surface of the installation notch **81**.

(45) It can be understood that the second matching surface **812** is perpendicular to the first matching surface **811**. More specifically, the perpendicular line of the first mating surface **811**

extends along a second direction, the perpendicular line of the second mating surface **812** extends along a first direction, and the second direction is perpendicular to the first direction. The shape of the display device is not specific, it may be a prism or a cylinder, no matter the shape of the display device is a prism or a cylinder, the second mating surface **812** is not a flat surface, and thus the first direction is not a specific direction, but is an axis direction perpendicular to the second mating surface **812** and the display device. The second direction is generally the y-direction shown in FIG. **4** and the first direction may be the x-direction shown in FIG. **4**.

(46) The display substrate **1** is arranged at a side of the first mating surface **811**, and the cover plate **6** is arranged at a side of the display substrate **1** away from the first mating surface **811**. The side surface of the display substrate **1** and the side surface of the cover plate **6** are opposite to the second mating surface **812**. The orthographic projection of the first mating surface **811** on the cover plate **6** at least partially overlaps the orthographic projection of the display substrate **1** on the cover plate **6**. It can be understood that the distance between the side surface of the display substrate **1** and the second matching surface **812** is smaller than the width of the first matching surface **811**.

(47) The display device further includes the supporting member **9**, one end of which is connected to a side surface of the cover plate **6**, and the other end of which is connected to the first mating surface **811**. The cover plate **6** and the middle frame **8** are connected together through the supporting member **9**. The supporting member **9** includes the first supporting section **91** and the second supporting section **92** connected sequentially. A fourth adhesive layer **82** is provided at a side of the first supporting section **91** close to the first mating surface **811**. The first supporting section **91** is adhered to the first mating surface **811** through the fourth adhesive layer **82**, and the second supporting section **92** is adhered to the side surface of the cover plate **6** through the fifth adhesive layer **63**.

(48) The first mating surface **811** may be set as a stepped shape, the first supporting section **91** may be set as a stepped shape matching the shape and size of the first mating surface **811**, and the stepped first mating surface **811** is matched with the stepped first supporting section **91**, which effectively increases the bonding area and the pressing contact area between the middle frame **8** and the supporting member **9**, and greatly enhances the connection stability between the middle frame **8** and the supporting member **9**.

(49) A first notch **61** may be provided at a position of the side surface of the cover plate **6** opposite to the second mating surface **812**. The first notch **61** extends along the thickness direction of the cover plate **6**. The second supporting section **92** is partly arranged in the first notch **61**. The fifth adhesive layer **63** may be set on the end surface and the side surface of the first notch **61**, and the second supporting section **92** is connected to the end surface and the side surface of the first notch **61** through the fifth adhesive layer **63**.

(50) It can be understood that the second supporting section **92** is arranged between the side surface of the first notch **61** and the second mating surface **812**. One surface of the second supporting section **92** is attached to the second mating surface **812**, and the other surface is partially attached to the side surface of the first notch **61**. Generally, the thickness of the second supporting section **92** is equal to the distance between the side surface of the first notch **61** and the side surface of the cover plate **6**. In order to prevent that a fitting gap exists between the cover plate **6** and the middle frame **8** and external dust enters the fitting gap, the side surface of the cover plate **6** is close to the second mating surface **812** in the thickness direction of the cover plate **6**.

(51) The smaller the distance between the side surface of the cover plate **6** and the second matching surface **812**, the better, and the ideal situation is that the distance between the side surface of the cover plate **6** and the second matching surface **812** is zero. A gap is formed between the side surface of the first notch **61** and the second mating surface **812**, and the second supporting section **92** is partly arranged in the gap between the first side surface and the second mating surface **812**, and the thickness of the second supporting section **92** is equal to the width of the gap.

(52) As shown in FIG. **5** and FIG. **6**, in order to prevent the connection between the second section

and the first notch **61** from becoming loose (which may result in peeling between the display panel and the middle frame **8**) when the display device is subjected to frequent vibrations, the side surface of the first notch **61** is also provided with a second notch **62**. The second notch **62** extends perpendicular to the thickness direction of the cover plate **6**. The supporting member **9** also includes a third supporting section **93**. The third supporting section **93** is connected to an end away from the first supporting section **91**. The third supporting section **93** is snap-fitted in the second notch **62**. Through the cooperation between the third supporting section **93** and the second notch **62**, the movement of the cover plate **6** in the thickness direction thereof can be restricted.

(53) However, the hardness and wear resistance of the cover plate **6** are usually smaller than those of the supporting member **9**. In order to prevent the supporting member **9** from directly contacting the cover plate **6** and affecting the strength of the cover plate **6**, the cross-sectional shape of the second notch **62** may be set as arc-shaped, and the cross-section of the third supporting section **93** is set as an arc shape matching the shape and size of the second notch **62**. When the display device is vibrated, the contact area between the first section and the second notch **62** may be increased, thereby making the cover plate **6** not easily damaged.

(54) A flexible buffer layer **94** may also be provided between the third supporting section **93** and the second notch **62**. The material of the flexible buffer layer **94** is usually an elastic material that deforms after being subjected to a force and returns to its original state after the force disappears, and specifically the material of the flexible buffer layer **94** may be foam. It can be understood that the foam is only an example of the flexible buffer layer **94**, and does not constitute a limitation to the flexible buffer layer **94**.

(55) The first supporting section **91** is connected to the first mating surface **811** and spaced apart from the display substrate **1**. Specifically, the display substrate **1** has a display region and a binding region located in the periphery of the display region. The orthographic projection of the first mating surface **811** on the cover plate **6** at least partially overlaps the orthographic projection of the binding region on the cover plate **6**. The supporting member **9** may play a certain protective role below the binding region, preventing the pressure generated by external impact during the assembly process from being transmitted to the binding region (which may result in poor display of the display panel).

(56) As shown in FIG. 7, in order to better protect the binding region of the display panel, the width of the first supporting section **91** may be increased. Specifically, the width of the first supporting section **91** is set to be greater than or equal to the width of the first mating surface **81**. It needs to be at least ensured that the orthographic projection of the binding region on the cover plate **6** is located within the orthographic projection of the first supporting section **91** on the cover plate **6**.

(57) In addition, the depth of the second notch **62** along the first direction should be as small as possible under the condition that the position of the third supporting section **93** in the second direction can be limited. The advantage of this arrangement is that, on the one hand, such arrangement can reduce the impact of the second notch **62** on the strength of the cover plate **6**, and on the other hand, such arrangement can prevent the second notch **62** from extending to the display region of the display panel (which will affect the display effect of the display panel).

(58) The depth of the second notch **62** is generally smaller than the width of the first mating surface **811**, while the width of the third supporting section **93** is usually equal to the depth of the second notch **62**, and the width of the first supporting section **91** is usually equal to the width of the first mating surface **811**. Therefore, the width of the third supporting section **93** is generally smaller than the width of the first supporting section **91**. It should be noted that the width of the first supporting section **91** here refers to the length of the first supporting section **91** extending along the first direction, the width of the third supporting section **93** refers to the length of the third supporting section **93** extending along the first direction, and the width of the first mating surface **811** refers to the length of the first mating surface **811** extending along the first direction.

(59) It should be noted that the thicknesses of the first supporting section **91**, the second supporting

section **92** and the third supporting section **93** of the supporting member **9** are generally equal. The supporting member **9** adopts a material which has good strength and may be processed into an ultra-thin structural member. For example, the material of the supporting member **9** is stainless steel.

(60) As shown in FIG. **8**, in order to facilitate the assembly of the supporting member **9** with the cover plate **6** and the middle frame **8**, the supporting member **9** may be divided into two parts along the extending direction of the first side surface. The two parts of the supporting member **9** are respectively connected to the cover plate **6** and the middle frame **8**. Especially, when the third supporting section **93** of the supporting member **9** is connected to the second notch **62**, the third supporting section **92** may be snapped into the second notch **62** without changing the relative position of the third supporting section **93** and the second supporting section **92**.

(61) The two parts of the supporting member **9** may be the same or not. For example, the supporting member **9** may include two identical supporting units **901**. When the supporting member **9** is connected to the cover plate **6** and the middle frame **8**, one of the supporting units **901** may be connected to the first notch **61**, the second notch **62** and the first notch **62**, and then the other one of the supporting units **901** may be connected to the first notch **61**, the second notch **62** and the first mating surface **811**.

(62) As shown in FIG. **9**, in other possible implementations, the installation notch **81** may be set as a groove-like structure, that is, the middle frame **8** is provided with a limiting portion that cooperates with a surface of the cover plate **6** away from the first mating surface **811**. The limiting portion prevents peeling of the display panel along the direction away from the first mating surface **811**. The second supporting section **92** may cooperate with the end surface of the first notch **61** to prevent the display panel from peeling along the direction close to the first matching surface **811**. It can be understood that the movement of the cover plate **6** in its thickness direction may be limited by the cooperation between the limiting portion and the second supporting section **92**.

(63) The second notch **62** extends continuously along the extending direction of the edge of the cover plate **6**, and the length of the third supporting section **93** along the first direction is the same and equal to the length of the second notch **62**. In other possible implementations, the second notch **62** includes a plurality of recesses, and the plurality of recesses are distributed at intervals along the extension direction of the side surface of the cover plate **6**. The third supporting section **93** includes a plurality of protrusions arranged at intervals. The size and shape of each of the protrusions match the size and shape of a corresponding recess, and each of the protrusions is arranged in a corresponding recess.

(64) The cover plate **6** may be a glass cover plate, the central portion of the cover plate **6** is flat, and the edge portion of the cover plate **6** is arc-shaped along the direction away from the central portion. The edge portion is bent towards the lower surface of the glass cover plate **6** along the direction away from the central portion. The middle frame **8** fits with the edge portion in a direction away from the central portion. It should be noted that the edge portion may be bent towards the lower surface of the cover plate **6** as a whole, or only the upper surface of the edge portion may be bent towards the lower surface of the cover plate **6**. In this embodiment, the upper surface of the cover plate **6** refers to the surface away from the display substrate **1**, which is also the surface directly faced by a user during usage. The lower surface of the cover plate **6** refers to the surface close to the display substrate **1**, i.e., the surface away from the user.

(65) As shown in FIG. **10**, in order to further fix the relative position between the display panel and the middle frame **8** in the second direction, a third notch **83** may be provided on the second mating surface **812**. The third notch **83** and the second notch **62** are in the same straight line. The supporting member **9** further includes a fourth supporting section **95**. The fourth supporting section **95** is snap-fitted in the third notch **83**. The first supporting section **91** cooperates with the first mating surface **811** to define the relative position of one end of the supporting member **9** and the middle frame **8**, and the fourth supporting section **95** cooperates with the third notch **83** to define

the relative position of the other end of the supporting member **9** and the middle frame **8**, so as to ensure that the supporting member **9** will not move relative to the middle frame **8** in the second direction. It can be understood that if the position of the supporting member **9** is fixed, the position of the cover plate **6** is naturally fixed.

(66) The middle frame **2** may be set as an integral structure or a spliced structure. When the middle frame **2** is set as an integral structure, it is needed to ensure that the supporting member **9** has a certain amount of deformation. After the supporting member **9** is connected to the cover plate **6**, the supporting member **9** is connected to the middle frame **8**. The fourth supporting section **95** is bent to be substantially parallel to the second supporting section **92**, and the supporting member **9** is slid in along the second mating surface **812**. When the fourth supporting section **95** is aligned with the third notch **83**, the fourth supporting section **95** is restored to be perpendicular to the second supporting section **92** and snapped into the third notch.

(67) It should be noted that, in addition to the middle frame and the display panel, the display device also includes other necessary components and elements, such as housing, circuit board(s), power line(s), and so on. One of ordinary skill in this art can appreciate other supplements according to specific usage requirements of the display device, and details will not be repeated here.

(68) The display device may be a traditional electronic device, such as a mobile phone, a watch, a computer, a TV or a video recorder, or an emerging wearable device, such as VR glasses, which are not listed here.

(69) As shown in FIG. **11** and FIG. **12**, display device may be set as circular, then the shape of the cover plate **6** of the display device is circular, the shape of the middle frame **8** of the display device is ring-shaped. The supporting member **9** of the display device **9** includes two half-ring shaped supporting units **901**. Or, the display device may be set as rectangle, then the shape of the cover plate **6** of the display device is rectangle, and the shape of the middle frame **8** of the display device is a rectangular frame. The supporting member **9** of the display device includes two supporting units **901** of semi-rectangular frame structures. When the shape of the display device is circular, the display device may be a watch or VR glasses, when the shape of the display device is rectangular, the display device may be a mobile phone or a computer.

(70) The present disclosure will be readily apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. The present disclosure is intended to cover any modification, use or adaptation of the present disclosure, and these modifications, uses or adaptations follow the general principles of the present disclosure and include common knowledge or conventional technical means in the technical field not disclosed in the present disclosure. The specification and embodiments are to be considered as exemplary only, with the true scope and spirit of the present disclosure indicated by the appended claims.

Claims

1. A display device, comprising: a middle frame provided with a first mating surface; a display panel comprising a display substrate and a cover plate arranged on a side of the display substrate, wherein a first notch is provided at a side of an edge of the cover plate close to the first mating surface, the first notch extends along a thickness direction of the cover plate, and an orthographic projection of the display substrate on the cover plate at least partially overlaps an orthographic projection of the first mating surface on the cover plate; and a supporting member comprising a first supporting section and a second supporting section which are connected sequentially, wherein the first supporting section is connected to the first mating surface, and the second supporting section is partially arranged in the first notch; wherein a second notch is provided at a side of the first notch, the second notch extends perpendicular to a thickness direction of the cover plate, and the supporting member further comprises a third supporting section which is connected to an end

away from the first supporting section, and the third supporting section is snap-fitted in the second notch; and wherein a third notch is provided at a second mating surface, and the supporting member further comprises a fourth supporting section, and the fourth supporting section is snap-fitted in the third notch.

2. The display device according to claim 1, wherein the middle frame further comprises the second mating surface, and a gap is formed between a side surface of the first notch and the second mating surface, and the second supporting section is partly arranged in the gap, and a thickness of the second supporting section is equal to a width of the gap.

3. The display device according to claim 1, wherein the first supporting section is spaced apart from the display substrate.

4. The display device according to claim 3, wherein the display substrate has a display region and a binding region located at a periphery of the display region, and the orthographic projection of the first mating surface on the cover plate at least partially overlaps an orthographic projection of the binding region on the cover plate.

5. The display device according to claim 4, wherein a width of the first supporting section is greater than or equal to a width of the first mating surface, and the orthographic projection of the binding region on the cover plate is located in an orthographic projection of the first supporting section on the cover plate.

6. The display device according to claim 1, wherein a width of the third supporting section is smaller than a width of the first supporting section.

7. The display device according to claim 1, wherein: the middle frame is provided with an installation notch, the cover plate is arranged in the installation notch, and the first mating surface is an end surface of the installation notch; and the middle frame further comprises the second mating surface, the second mating surface is a side surface of the installation notch, and the first notch and the second notch are arranged at a position of the side surface of the cover plate which is opposite to the second mating surface.

8. The display device according to claim 1, wherein the supporting member is attached to a side surface of the first notch and divided into at least two parts along an extending direction of the side surface.

9. The display device according to claim 1, wherein a flexible buffer layer is provided between the third supporting section and the second notch.

10. The display device according to claim 1, wherein the supporting member is made of stainless steel.

11. The display device according to claim 1, wherein the second notch extends continuously along an extending direction of an edge of the cover plate.

12. The display device according to claim 1, wherein the second notch comprises a plurality of recesses, and the plurality of recesses are distributed at intervals along the extending direction of the side surface of the cover plate.

13. The display device according to claim 1, wherein the edge of the cover plate is arc-shaped, and a height of the edge of the cover plate away from the first mating surface is gradually decreased from an end close to a center of the cover plate to an end away from the center.

14. The display device according to claim 1, wherein the first supporting section is adhered to the first mating surface.

15. The display device according to claim 1, wherein the third notch and the second notch are located in a same straight line.

16. The display device according to claim 1, wherein a limiting portion is arranged on the middle frame to prevent the display panel from moving along a direction away from the first mating surface.
